

36 - Organic synthesis

Identification tests

alcohols

- ★ oxidation with acidified KMnO_4
- ★ result: orange to green

carboxylic acids

- ★ reaction with carbonates
- ★ results: produces CO_2 (effervescence)

esters

- ★ hydrolysis to yield alcohols & carboxylic acids

halides

- ★ AgNO_3 test (AgNO_3 in ethanol & heat)
- ★ results: $\text{F}^- \rightarrow$ no ppt. , $\text{Cl}^- \rightarrow$ white ppt. ,
 $\text{Br}^- \rightarrow$ pale ppt. , $\text{I}^- \rightarrow$ yellow ppt.

aldehydes

- ★ Tollens' reagent or Fehling's solution
- ★ results: silver mirror or blue to red ppt.

ketones

- ★ negative results with Tollen's & Fehling's
- ★ positive result with 2,4-DNPH
- ★ results: orange ppt.

amines

- ★ react with HCl to form ammonium salt
- ★ ammonium salt produces NH_3 if heated
- ★ results: NH_3 turns damp red litmus paper $\rightarrow$ blue

nitriles

- ★ hydrolysis to form carboxylic acids or amines

acyl chlorides

- ★ AgNO_3 test
- ★ results: white ppt of AgCl

benzene

- ★ bromination test
- ★ w/o catalyst, benzene cannot react with Br_2
- ★ results: bromine remains orange-brown (no change)

phenol

- ★ reaction with neutral FeCl_3
- ★ results: violet/purple complex forms